

Shi Wang

Ph.D. Candidate, Geodynamics/Geophysics

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Profile

I am a geodynamics researcher specializing in numerical simulation of lithosphere deformation, mantle convection, and petrological modelling of mantle melting. I developed a coupled framework that integrates peridotite melting petrological model into thermomechanical simulations, enabling the quantitative prediction of magma production and chemical evolution. My doctoral research applies this approach to constrain the evolution of lithospheric rifting and breakup.

My methodology centers on bridging the gap between physical geodynamics and observable surface data. I am particularly interested in extending my approach to a broader range of tectonic regimes, including intra-plate and plume-related magmatism. I am also expanding my focus on the coupling of surface and deep Earth processes. Building on my experience linking surface uplift histories to mantle dynamics, I aim to further investigate how surface vertical motions not only record deep mantle processes, but may also modulates the of mantle flow and related magmatism.

Education

September 2019 – **Ph.D. Candidate** – nearing completion, School of Earth and Space Sciences, University of Science and Technology of China

Present: **Dissertation in preparation:** *Magmatism in Lithosphere Rift and Breakup: Thermomechanical-petrological modelling and implications*

Developed a coupled geodynamic–geochemical framework to predict peridotite melting and its major-element chemistry in upper mantle dynamics. Applied to lithospheric rifting and breakup, this work quantifies how mantle temperature, source composition, rift kinematics, and lithosphere rheology control magma production and chemistry.

Supervisor: Wei Leng, University of Science and Technology of China

September 2015 – **Bachelor's Degree in Geophysics**, School of the Gifted Young, University of Science and Technology of China

June 2019 **Thesis:** *Numerical simulation of Mid-Ocean ridge migration and Discussion about its Physical Mechanism*

Investigated ridge migration under varying plate spreading conditions using analytical and numerical modeling.

Supervisor: Wei Leng, University of Science and Technology of China

Publications

Published:

- **Wang, S.**, Leng, W. Breakup of strong cratonic lithosphere causes extensive magmatism at continental margins. *Sci Rep* **16**, 12978 (2026). <https://doi.org/10.1038/s41598-026-42222-7>
- Leng, W., **Wang, S.**, Liu, H. Mantle plume-lithosphere interaction: Evolution and resource environmental effects. *Sci. China Earth Sci.* (2026). <https://doi.org/10.1007/s11430-025-1842-0>
Review article. I am responsible for part of the writing and review.

In preparation:

- **Wang, S.**, Leng, W. Evolution of major element chemistry during lithosphere rifting – a coupled geochemical-geodynamic modelling. *Expected submission June 2026*.
- Su, H., Leng, W., **Wang, S.**, Gerya, T. Magmatic Segmentation During Rifting and Its Implications for the Timing of Seafloor Spreading in the Red Sea.
I am responsible for forward gravity modelling and related discussion.

Conferences:

- May., 2026, EGU General Assembly, Vienna, Austria
Poster: *Breakup of strong cratonic lithosphere causes extensive magmatism by rift shoulder subsidence*, <https://doi.org/10.5194/egusphere-egu26-5200>
- Oct., 2023, Annual Meeting of Chinese Geoscience Union (CGU), Zhuhai, China
Oral presentation: *Magmatism chemistry in continental rifting and the influence of plume*

Work experience

2023-2025 **Teaching assistant** for *Geodynamics* inducted by Prof. Wei Leng.

2019-2020 **Writer and assistant editor** for two popular science platforms supported by USTC: *Dr. Stone* focus on geoscience and *First Light* focus on astronomy.

Referee

Prof. Wei Leng, School of Earth and Space Sciences, University of Science and Technology of China, wleng@ustc.edu.cn